

Week-5

OBJECTIVE: Learn the basics of Apache Spark and use it to clean, transform and analyze data using DataFrames.

Steps to perform:

Step 1: Open Google Collab and rename it to spark_basics.ipynb

Step 2: Install PySpark using !pip install pyspark

Step 3: Import Libraries

```
from pyspark.sql import SparkSession  
from pyspark.sql.functions import *  
from pyspark.sql.types import *
```

Step 4: Create Spark Session

```
spark = SparkSession.builder.appName("Week5Assignment").getOrCreate()
```

ASSIGNMENT QUESTION-ANSWER

Q1: What are the key limitations of traditional MapReduce that make Spark a preferred choice for modern big data processing?

Ans: Traditional MapReduce suffers from:

- Disk-based processing causing high latency.
- Repeated read/write operations.
- Inefficient iterative computations.
- Complex programming model.

Spark is preferred because:

- It uses in-memory computation.
- Provides faster execution.
- Supports SQL, Machine Learning, and Streaming.
- Offers easy-to-use DataFrame APIs

Q2: Explain how Spark uses In-Memory Computing to speed up iterative machine learning algorithms compared to disk-based systems.

Ans- Spark stores intermediate data in memory instead of writing it to disk after each step. This significantly reduces disk I/O and speeds up iterative machine learning algorithms.

Q3: Write a code snippet to remove all duplicate rows from a DataFrame based on a specific set of columns: user_id and transaction_date.

Ans- `df = df.dropDuplicates(["user_id", "transaction_date"])`

Q4: Given a DataFrame df_sales, write a query to filter for rows where the region is 'West' and then group by product_category to find the average sale_amount.

Ans-

```
df_sales.filter(
    col("region") == "West"
).groupBy(
    "product_category"
).agg(
    avg("sale_amount").alias("Average_Sale")
).show()
```

```
... +-----+-----+
|product_category|Average_Sale|
+-----+-----+
|      Electronics|533.333333333334|
|      Clothing|250.0|
+-----+-----+
```

Q5: What is the difference between .na.drop() and .na.fill()? Provide a code example of filling null values in a status column with the string 'Unknown'.

Ans- `.na.drop()` removes rows containing null values.

`.na.fill()` replaces null values with specified values.

```
df = df.na.fill(
    "Unknown",
    subset=["status"]
)
```

Q6: Write a query to find the total count of records for each city in a DataFrame, but only for cities where the count is greater than 100.

Ans-

```
df.groupBy(
  "city"
).count().filter(
  col("count") > 100
).show()
```

```
*** +-----+-----+
    |      city|count|
    +-----+-----+
    |   Phoenix|    1|
    |Los Angeles|    2|
    |   Chicago|    1|
    |   Houston|    2|
    |   New York|    1|
    +-----+-----+
```

Q7: How does the immutability of Spark DataFrames affect how you perform "data cleaning" steps like dropping columns or renaming them?

Ans- Spark DataFrames are immutable. Operations such as dropping columns or renaming columns create a new DataFrame instead of modifying the original DataFrame.

Example:

```
new_df = df.drop("age")
```

```
new_df = df.withColumnRenamed(
  "price",
  "sale_price"
)
```

Q8: Write a Spark command to filter a dataset for rows where the age is between 18 and 30 (inclusive) and the subscription is 'Premium'.

Ans-

```
df.filter(
  (col("age") >= 18) &
  (col("age") <= 30) &
  (col("subscription") == "Premium")
)
```

).show()

...	user_id	transaction_date	name	age	subscription	region	product_category	sale_amount	status	city	email	username	price	store_id	event_time
	101	2025-08-01	John	25	Premium	West	Electronics	500	Completed	New York	john@gmail.com	john123	500	1	2025-08-01 10:30:00
	103	2025-08-03	Bob	22	Premium	West	Electronics	700	Completed	Los Angeles	bob@gmail.com	bob789	700	1	2025-08-03 09:45:00
	104	2025-08-04	David	28	Premium	South	Furniture	450	Pending	Houston	david@gmail.com	david123	450	3	2025-08-04 14:20:00
	108	2025-08-08	Mike	18	Premium	West	Electronics	400	Pending	Los Angeles	mike@gmail.com	mike999	400	2	2025-08-08 15:00:00

Q9: When cleaning a dataset, why is it often better to handle null values before performing mathematical aggregations like sum() or avg()?

Ans-

- Null values can lead to inaccurate or misleading results during aggregation.
- Missing values may cause incorrect calculations of average, sum, or other statistics.
- Replacing or removing null values improves data quality and reliability.
- Handling null values ensures consistency in analytical results.
- It helps avoid unexpected errors during data processing.
- Clean data produces more accurate insights and better decision-making.

Q10: Write the code to revise a column named raw_timestamp by casting it to a TimestampType and renaming it to event_time.

Ans-

```
df = df.withColumn(
    "event_time",
    col("raw_timestamp").cast(TimestampType())
).drop("raw_timestamp")
```

 <code>df.select("event_time").show(truncate=False)</code>	<code>df.printSchema()</code>
<pre>... +-----+ event_time +-----+ 2025-08-01 10:30:00 2025-08-02 11:15:00 2025-08-03 09:45:00 2025-08-04 14:20:00 2025-08-05 08:30:00 2025-08-08 15:00:00 2025-08-09 16:20:00 +-----+</pre>	<pre>root -- user_id: integer (nullable = true) -- transaction_date: date (nullable = true) -- name: string (nullable = true) -- age: integer (nullable = true) -- subscription: string (nullable = true) -- region: string (nullable = true) -- product_category: string (nullable = true) -- sale_amount: integer (nullable = true) -- status: string (nullable = false) -- city: string (nullable = true) -- email: string (nullable = true) -- username: string (nullable = true) -- price: integer (nullable = true) -- store_id: integer (nullable = true) -- event_time: timestamp (nullable = true)</pre>

Q11: Explain the "Shuffle" process that occurs during a grouping operation. Why is it considered a wide transformation?

Ans-

- Shuffle is the process of redistributing data across different partitions in a cluster.

- ii. Operations like groupBy(), join(), and reduceByKey() require shuffling.
- iii. During shuffle, Spark exchanges data between executors to bring similar keys together.
- iv. Shuffle involves network communication and disk I/O, making it expensive.
- v. It increases execution time and resource consumption.
- vi. It is called a **wide transformation** because data from multiple partitions is combined and redistributed across partitions.

Q12: Write a code snippet that identifies and removes rows where the email column contains null values OR the username is an empty string.

Ans-

```
df = df.filter(
    col("email").isNotNull() &
    (col("username") != "")
)
```

Q13: How do you use the .agg() function to calculate multiple statistics at once, such as the min, max, and mean of the price column?

Ans-

```
df.agg(
    min("price").alias("Minimum"),
    max("price").alias("Maximum"),
    mean("price").alias("Average")
).show()
```

```
|-----+-----+-----+
|Minimum|Maximum|          Average|
|-----+-----+-----+
|      0|    700|385.7142857142857|
|-----+-----+-----+
```

Q14: In the context of cleaning a dataset, what is the risk of using inferSchema=true when your source data contains messy or inconsistent date formats?

Ans-

user_id	transaction_date	name	age	subscription	region	product_category	sale_amount	status	city	email	username	price	store_id	raw_timestamp
101	2025-08-01	John	25	Premium	West	Electronics	500	Completed	New York	john@gmail.com	john123	500	1	2025-08-01 10:30:00
102	2025-08-02	Alice	30	Basic	East	Clothing	300	NULL	Chicago	alice@gmail.com	alice456	300	2	2025-08-02 11:15:00
103	2025-08-03	Bob	22	Premium	West	Electronics	700	Completed	Los Angeles	bob@gmail.com	bob789	700	1	2025-08-03 09:45:00
104	2025-08-04	David	28	Premium	South	Furniture	450	Pending	Houston	david@gmail.com	david123	450	3	2025-08-04 14:20:00
105	2025-08-05	Eva	35	Basic	West	Clothing	250	NULL	Phoenix	eva@gmail.com	eva567	NULL	2	2025-08-05 08:30:00
101	2025-08-01	John	25	Premium	West	Electronics	500	Completed	New York	john@gmail.com	john123	500	1	2025-08-01 10:30:00
106	2025-08-06	Sam	19	Premium	West	Electronics	600	Completed	Chicago	NULL	sam123	600	1	2025-08-06 12:45:00
107	2025-08-07	Lisa	27	Premium	East	Furniture	550	Completed	New York	lisa@gmail.com	NULL	550	3	2025-08-07 13:00:00
108	2025-08-08	Mike	18	Premium	West	Electronics	400	Pending	Los Angeles	mike@gmail.com	mike999	400	2	2025-08-08 15:00:00
109	2025-08-09	Sara	29	Basic	South	Clothing	350	Completed	Houston	sara@gmail.com	sara111	350	2	2025-08-09 16:20:00

Q15: Write a final processing pipeline that:

1. Filters out duplicates.
2. Fills null prices with 0.
3. Groups by store_id to calculate total revenue.

Ans-

```
# Remove duplicates
```

```
df1 = df.dropDuplicates()
```

```
# Fill null prices with zero
```

```
df2 = df1.na.fill(
```

```
0,
```

```
subset=["price"]
```

```
)
```

```
# Calculate total revenue
```

```
result = df2.groupBy(
```

```
"store_id"
```

```
).agg(
```

```
sum("price").alias("total_revenue")
```

```
)
```

```
result.show()
```

store_id	total_revenue
1	1200
3	450
2	1050